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Inclined Pullout Capacity of Suction Anchors in Clay over Silty Sand

Mi Zhou[1]; Ningxin Yang[2]; Yinghui Tian[3]; and Xihong Zhang[4]

Abstract: This study presents a range of numerical modeling to evaluate the pullout capability of suction anchors subjected to inclined mooring loading in a clay-over-silty-sand deposit. Comprehensive numerical models were generated, which were validated against existing centrifuge testing and numerical modeling results. A parametric study was then carried out to assess the effects of the anchor length, padeye location, mooring angle, and layered deposits on the pullout capacity of suction anchors. According to the numerical simulation results, the following key design factors were obtained: (1) the optimal padeye position, (2) the ultimate horizontal and vertical pullout abilities, and (3) the horizontal load distributions at different padeyes. A simple framework of calculating the combined VH surfaces of the anchors was developed for preliminary prediction of the pullout capacity of suction anchors in a clay-over-silty-sand deposit that provides the optimized location of padeyes and more accurate prediction for the holding capacity of anchor. The design method proposed by this study can contribute to optimize and improve the current design guideline for more accurately predicting the bearing capacity of anchors in clay-over-silty-sand soil deposits. **DOI: 10.1061/JGGEFK.GTENG-11074.** *© 2023 American Society of Civil Engineers.*

Author keywords: Suction anchor; Clay over silty sand; Inclined loading; Finite element; Pullout capacity.

Introduction

Suction anchors are commonly utilized in coastal floating platforms and deepwater projects, such as pipeline manifolds or floating ships (Andersen and Jostad 1999; Tjelta 2001; Zhu et al. 2014; Fu et al. 2020; Cheng et al. 2021; Mohiuddin et al. 2022; Al-Janabi and Aubeny 2022). In general, a suction anchor consists of a top cap, a mooring chain, and a peripheral skirt as shown in Fig. 1(a). At the initial installation stage, the skirt is forced into the seafloor by self-weight. Water is then pumped out, creating a pressure difference between the soil and the skirt, leading to the further penetration of the anchor (Erbrich and Tjelta 1999; Houlsby and Byrne 2005; Andersen et al. 2008; Senders and Randolph 2009; Zhao et al. 2019; Al-Janabi and Aubeny 2022). When the top lid is near to the seafloor, the penetration procedure is finished. In the operation phase, as shown in Fig. 1(b), because of the actions of waves, wind, and current, floating structures need catenary and taut mooring chains that can impose an inclined load on the suction anchor. Thus, suction anchors are naturally subjected to combined loading conditions.

During the design process, the pullout capacity of anchors is one of the main concerns and has been paid great attention in engineering applications. The ultimate holding capacity of a suction anchor strongly depends on the inclined loading angles, anchor length, loading point [called padeye, shown in Fig. 1(a)], and soil conditions. Different loading conditions could greatly influence the pullout capacities because the failure mechanism would vary from rotational to translational movement (Bang et al. 2006, 2011). It is therefore necessary to accurately predict the ultimate pullout performance of suction anchors under complex loading scenarios.

A number of studies, including theoretical research, numerical methods, and experimental testing, have been carried out to investigate the ultimate bearing capacities of suction anchors under inclined loading. Some of existing literatures are focused on anchors in either pure clay (e.g., Randolph and House 2002; Clukey et al. 2004) or sandy deposits including silica sand and silty sand (Bang et al. 2011; Kim et al. 2015, 2016; Zhao et al. 2019). The performance of an anchor in silty sand has been paid much attention to address the pullout resistance of a single suction anchor and a group of suction anchors (Kim et al. 2015, 2016). It was found that the behavior of an anchor in the soil profile involved with silty sand is significantly different from that in silica sand. However, a pure sand soil profile rarely exists (Zhou et al. 2016; Ma et al. 2016), and most of the conditions in the Yellow Sea encountered in offshore projects are a layer of soil deposit involving one layer of silty sand, i.e., a soil profile of clay overlying silty sand. For other typical layered deposits, existing studies accounted for stiff-over-soft-clay, clay-over-sand, or sand-over-clay cases. Xia et al. (2021) investigated the in-place bearing capacity of suction caissons in stiff over soft clay. Zou et al. (2018) investigated the bearing behaviors of suction caissons in sand over clay. Ragni et al. (2020) investigated the influence of a top layer of clay on the performance of a suction caisson installed in sand.

Despite of these studies, the behavior of suction anchors in layered deposits, especially in stratified strata with silty sand is still not fully understood. For example, in some situations as shown in Fig. 1(b), clay over silty sand is prevalent in petroleum regions such as the Yellow Sea or western coastline of Korea. A limited amount

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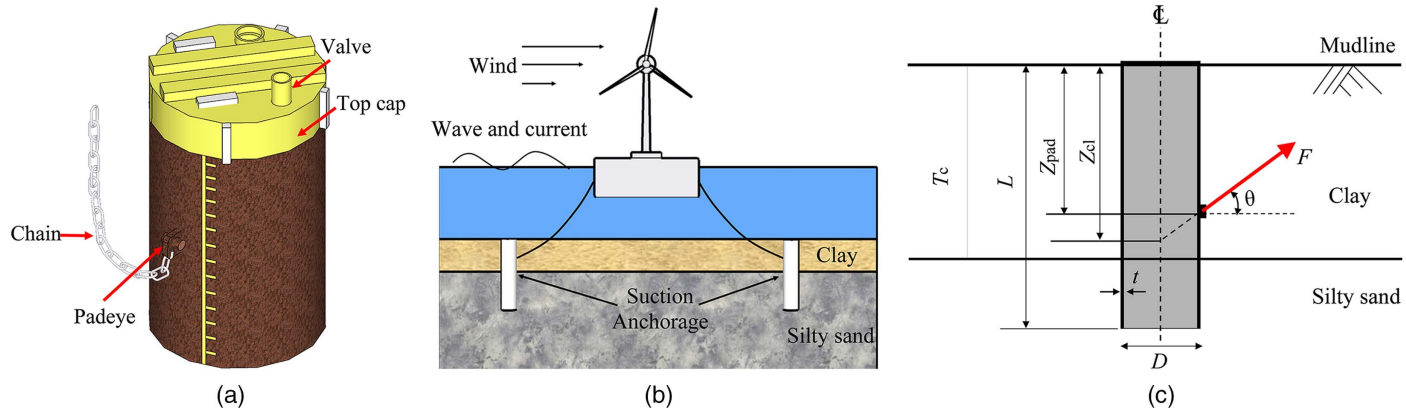

Fig. 1. (a) Illustration of suction anchor; (b) schematic view of a suction anchor in clay over silty sand; and (c) diagram of suction anchor subjected to inclined loading.

of research is available in the literature on the behavior of suction anchors in clay over silty sand, which limits the development with such a soft clay seabed. A higher stability risk of suction anchors was expected from the uncertainties in estimating the pullout capacities in such layered deposits. Current design guides such as American Petroleum Institute (API 2011) and DNV (Bhattacharya 2019) usually follow the guidelines of caisson in sand or in clay. However, it was proven that the behavior of a structure in sand is significantly different to that in silty sand, and the characteristics of the pile or shallow foundation in a layered soil profile is totally different from that in a single layer of sand or clay (Kim et al. 2015, 2016). Existing guidelines for designing piles in layered deposits mainly use the p-y curve method considering the equivalent friction coefficient and skin friction along the shaft. This method is not suitable for predicting the pullout capacity of suction anchors in layered deposits. The adopted method in existing guidelines can be unconservative for layered soil conditions, especially when a silty sand is overlain by soft clay. Therefore, a gap exists in the present design method and the true behavior of a caisson in a clay-over-silty-sand soil deposit. Insight into the behavior of caisson in clay-over-silty-sand soil is urgently needed.

This study targets the in-place bearing capacity of suction anchors in clay over silty sand. Numerical analysis was adopted to evaluate the behavior of suction anchors under inclined loading conditions. The following aspects were investigated: (1) design of padeye where the optimal attachment point is to be identified, (2) prediction of the ultimate (horizontal and vertical) pullout capacities of suction anchors, (3) determination of the horizontal pullout capacity of suction anchors at padeyes, and (4) quantification of the influence factors on the failure envelopes of suction anchors under inclined loading, and proposal of a practical design approach to assess the failure envelopes. The findings of this study can provide direct application for determining the optimized attachment point for improved pullout capacity of suction anchors in clay over silty sand. It establishes a design method to accurately predict the pullout capacity of anchors.

Numerical Modeling

Geometry and Mesh

A circular suction anchor with a diameter D and an embedment length L is considered, which is installed in clay-over-silty-sand deposits as shown in Fig. 1(c). The thickness of the overlying clay is denoted as T_c. For practical engineering application (Fu et al. 2020; Cheng et al. 2021; Al-Janabi and Aubeny 2022), the anchor's diameter D is set as 5 m, and the overlying clay thickness ratio T_c/D varies in the range of 1.0–3.0. Considering the mega size of floating structures, large anchors are investigated, which are to provide a large holding capacity (the length of a suction anchor can reach more than 30 m). Accordingly, the skirt length ratio L/D ranges from 1.5 to 6, and the thickness of the anchor skirt t is set as 50 mm. Considering the stiffness of steel anchors and soils, it is commonly considered as a rigid body to improve computational efficiency. This is a common practice in offshore geotechnics (Gourvenec 2008; Zhang et al. 2011; Zou et al. 2018; Xia et al. 2022). For the inclined loading on the anchor, the padeye position Z_{pad} is measured from the seabed to the attachment loading point, and the depth of the intersection between the centerline of the anchor and the inclined load F is defined as Z_{cl}. The inclined load F is applied to the padeye ranging from $0.3L$ to $0.9L$. Detailed analysis and parameters adopted can be found in Table 1.

Fig. 2 shows the boundary constraints and semicylindrical mesh. First-order and full-integration elements (C3D8) are employed. The degrees at the soil region base are fixed and the mesh

Table 1. Analysis details and parameters in the numerical analysis

Group	s_{um}/kD	$c_u/\gamma' D$	T_c/D	L/D	Note
Group 1	—	—	—	—	Model validation
Group 2	0, 1	0.07, 0.34	3.0	1.5–6.0	To investigate the effect of soil properties on the optimum centerline loading position $Z_{cl,opt}$
Group 3	0–1	0.07–0.34	1.0–3.0	1.5–6.0	To calculate the optimum centerline loading depth $Z_{cl,opt}$
Group 4	0–1	0.07–0.34	1.0–3.0	1.5–6.0	To calculate the ultimate uniaxial pullout capacity (H_{ult} and V_{ult})
Group 5	0–1	0.07–0.34	1.0–3.0	1.5–6.0	To calculate the horizontal pullout capacity distribution at different padeye positions
Group 6	0–1	0.07–0.34	1.0–3.0	1.5–6.0	To calculate VH failure envelopes

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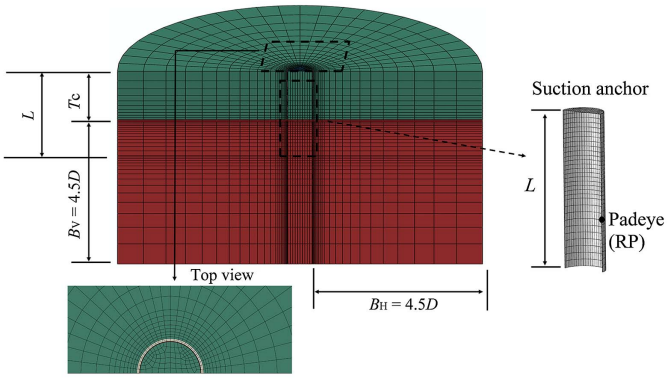

Fig. 2. Typical mesh for soil and suction anchor.

at the boundary are roller-constrained. The sensitivity investigation of mesh size was carried out and the soil element size near the anchor was set as $0.02D$. In Fig. 2, B_V and B_H are the vertical and horizontal distances measuring from the anchor to the boundary of the soil region, respectively. The ratio $B_H/D = B_V/D = 4.5$ is large enough to eliminate the influence of boundary effect compared to the size of the anchors (Hung and Kim 2014; Zou et al. 2018). The inclined load is directly applied at the padeye, where the moment on the anchor is ignored. Therefore, the padeye position can be considered as the load reference point (RP as shown Fig. 2).

Soil Properties and Interface Modeling

The overlying clay is modeled as an elastoplastic material that follows the Tresca yield rule, with associated flow rule. Poisson's ratio $\upsilon = 0.49$ is adopted to model the no-volume-change undrained conditions. Normally, the undrained shear strength of clay s_u shows the variation with depth as

$$s_u = s_{um} + kz \tag{1}$$

where s_{um} = undrained shear strength at the seabed; z = submerged depth below the seabed; and k = clay strength gradient with depth. A constant stiffness ratio $E/s_u = 500$ is adopted throughout the clay profile (where E is Young's modulus). In assessing the influence of soil properties on the pullout capacity, the normalized parameter s_{um}/kD is commonly used to define the strength variation of the normally consolidated soil. With references to previous studies (Davis and Booker 1973; Menzies and Roper 2008; Menzies and Lopez 2011; Hung and Kim 2014), s_{um}/kD varies as 0, 0.4, 0.67, and 1.0. More details about clay properties can be found in Table 2.

According to the Unified Soil Classification System (USCS) (ASTM 2017), the silty sand comprises a soil mixture with coarse and fine grains, usually consisting of clean sand and more than 12% silts (also called sand-silt mixtures). Experimental observations showed that the existence of fine silt can affect the undrained shear strength significantly, depending on the content of fine silt (Naeini and Mahdavi 2009; Chang and Yin 2011; Jin et al. 2013). Thus, the undrained shear strength of the silty sand is strongly related to the amount of the silt content. An abundance of data are available on the silty sand from the western coastline of Korea and in the Yellow Sea (Liu et al. 2014; Kim et al. 2015, 2016; Jia et al. 2018). Some collected typical particle size distributions of the silty sand samples are displayed in Fig. 3. In this study, the silty sand was modeled with a Mohr-Coulomb model. Considering different silt contents in the silty sand, the cohesion of the silty sand c_u varies from 3 to 15 kPa, where the normalized shear strength $c_u/\gamma' D$ is 0.07–0.34.

Table 2. Clay properties

Parameter	Profile			
	Clay 1	Clay 2	Clay 3	Clay 4
s_{um} (kPa)	0.1	3	5	10
k (kPa/m)	1.25	1.5	1.5	2
γ' (kN/m^3)	6	6	6	6
s_{um}/kD	0	0.4	0.67	1.0

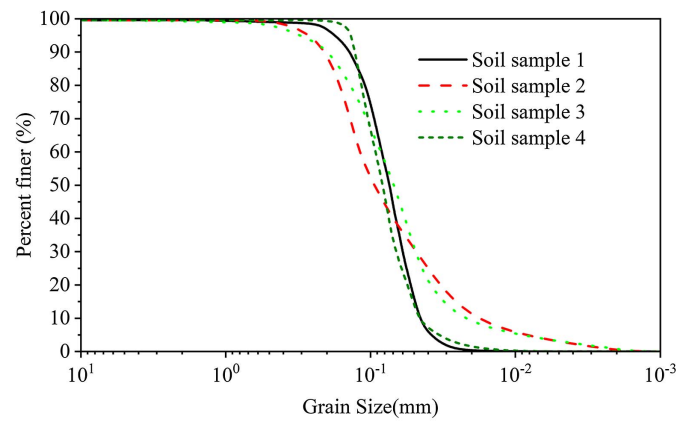

Fig. 3. Particle size distribution of silty sand samples. (Data from Kim et al. 2016.)

Table 3. Silty sand properties

Parameter	Profile			
	Sand 1	Sand 2	Sand 3	Sand 4
c_u (kPa)	3	6	10	15
ϕ (degrees)	33	33	33	33
ψ (degrees)	8	8	8	8
γ' (kN/m^3)	8.9	8.9	8.9	8.9
$c_u/\gamma' D$	0.07	0.13	0.22	0.34

Friction and dilation angles in all numerical analysis remain constant ($\phi = 33°$, $\psi = 8°$). Because the ultimate pullout capacity primarily relies on the soil strength, Young's modulus E is set as high enough (over $2{,}000c_u$). More details about the silty sand properties can be found in Table 3.

An undrained soil condition is assumed for all analyses in this study for offshore applications. For the silty sand, because the loading rate is high and the drainage path is long for large anchors, outflow of pore water is eliminated (Tapper 2013). Moreover, the existence of the overlying clay layer will also block the seepage path of the silty sand layer. Therefore, total stress analysis is employed to model the undrained condition of the clay and the silty sand.

A gap is assumed to exist between the external surface of the anchor and the soil following a previous study (Jeanjean et al. 2006), showing no indication of gapping condition. For the interface between the anchor and the clay, there is a lack of consensus on the formation of a gap despite a previous study by Fu et al. (2020) that showed that the strength intercept at the seabed is critical to the gap formation. For example, with a gap interface, gapping condition will not occur in the case of $s_{um}/kD = 0$, which nevertheless is likely to form in the case of $s_{um}/kD = 0.34$. Therefore, the gap interface is involved, and the formation of gaps is

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calculated automatically in the ABAQUS program calculation. This assumption is considered to be conservative and reasonable because, if a gap occurs, the suction force in the rear side of the anchor is neglected. During the analysis, the soil–skirt interface follows the Coulomb friction law where separation is allowed. The friction coefficient can be expressed as μ, where $\mu = 1$ for the clay and $\mu = 0.4$ for the silty sand, respectively.

Analysis Details and Loading Methods

Numerical modeling was conducted with the commercial software ABAQUS version 6.13. The anchor was modeled as wished in place (i.e., the installation process is not involved). Small-strain finite-element (FE) analysis was carried out. The FE modeling procedure includes three steps:

1. The whole soil domain achieves the initial equilibrium state. At this stage, the suction anchor is simulated as a void.
2. The suction anchor is activated using the model change method provided by ABAQUS. The interface between the suction anchor and soil is initiated at the instance. The model change method in ABAQUS enables removal or reactivation of elements or contact pairs during analysis to achieve the construction process of a suction anchor.
3. The suction anchor is then loaded at the padeye with an inclined loading. To obtain the pullout capacity of the suction anchor, a prescribed displacement is applied at the padeye numerically.

The tangent method is commonly used to obtain the uniaxial pullout capacity of the suction anchor. As illustrated in Fig. 4(a), when the load on the load–displacement curve converges and becomes stable, it indicates the soil reaches the limit equilibrium state, and the load is identified as the ultimate capacity of the anchor (V_0 denotes the uniaxial vertical pullout capacity; H_0 denotes the uniaxial horizontal pullout capacity). The dimensionless horizontal and vertical pullout capacity factors $N_{ch} = H_0/\gamma' D^3$ and $N_{cv} = V_0/\gamma' D^3$ are derived using the corresponding pullout capacity divided by the anchor diameter D and the effective unit weight γ'. In some special cases, when the anchor translates horizontally or vertically without rotation, the uniaxial pullout capacity can be defined as the ultimate uniaxial pullout capacity (H_{ult} and V_{ult}).

Fig. 4(b) shows the establishment of the VH envelopes, in which probe tests are carried out with the loading control method in a local coordinate system (local CSYS). With the loading method in a local CSYS proposed by Tian et al. (2016), probe tests can enable a clear observation on the failure envelopes. Through applying displacement along the major axes in the local CSYS, failure envelopes in the VH space can be easily uniquely linked and identified. Moreover, due to the coupled interaction between the horizontal and the vertical loads, the VH envelopes will show a protruding increase on the vertical pullout capacity. Accordingly, the maximum vertical pullout capacity is denoted as V_{max}.

When the anchor moves horizontally without rotation under the inclined loading, the loading point can be defined as the optimum padeye position $Z_{pad,opt}$. To obtain $Z_{pad,opt}$, in the numerical modeling, a prescribed displacement is applied to the loading point (RP) of the anchor without rotation. As demonstrated in Fig. 4(c), the padeye under the inclined loading F can be simplified as a vertical force V, a horizontal force H, and a constraining moment M. The padeye depth can be moved to a position where the moment is zero. And the optimal loading position along the anchor $Z_{pad,opt}$ can be obtained from the simple equilibrium. The location of the optimal padeye can be determined by dividing the reaction moment at the RP with the total horizontal force. When the inclined loading intersects the centerline of the anchor, this point is defined as $Z_{cl,opt}$. For a given loading angle θ, when the anchor moves pure horizontally without rotation, $Z_{cl,opt}$ can be obtained by simply transforming

$$Z_{pad,opt} = \frac{M}{H} \tag{2}$$

$$Z_{cl,opt} = Z_{pad,opt} + \left(\frac{D}{2}\right) \times \tan(\theta) \tag{3}$$

When the anchor is loaded horizontally (the loading angle $\theta = 0°$), $Z_{pad,opt}$ and $Z_{cl,opt}$ can be regarded as the same value.

Model Validation

To validate the accuracy of the numerical model, simulation results for a suction anchor under inclined loading were compared with existing FE results and centrifuge testing data. Because there are no data on the behavior of suction anchors in layered deposits, model validation was performed with anchors in normally consolidated (NC) clay and silty sand separately.

For suction anchors in NC clay, a recent study on the deepwater anchor design practice sponsored by the American Petroleum Institute (API project) was included for comparison. Numerical

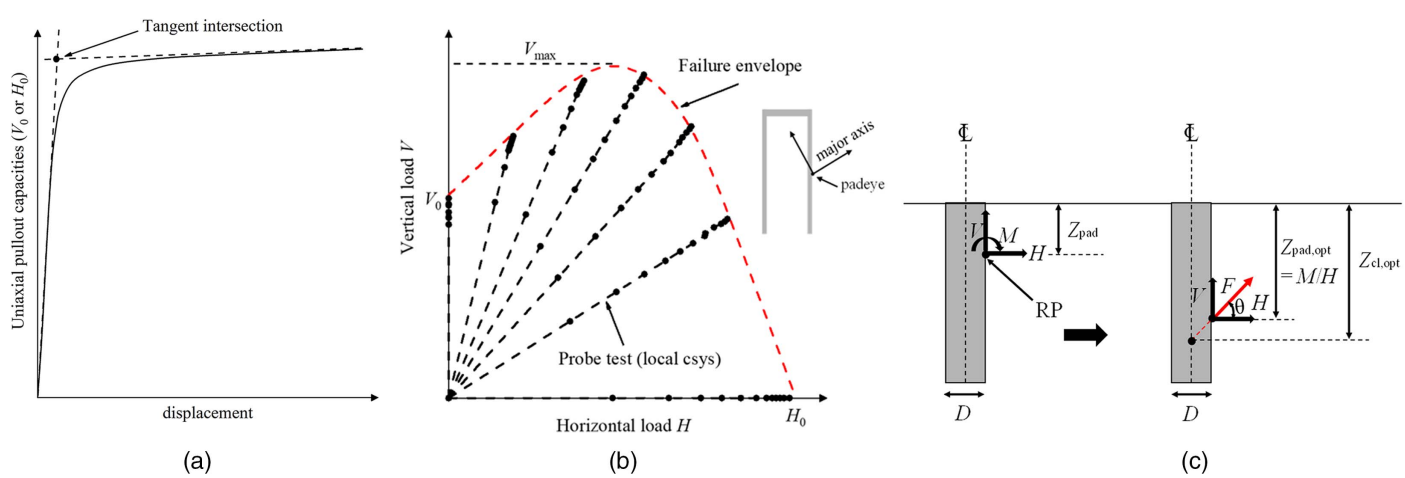

Fig. 4. (a) Determination of the uniaxial pullout capacity; (b) determination of the combined VH failure envelopes; and (c) determination of the optimum padeye and centerline depth.

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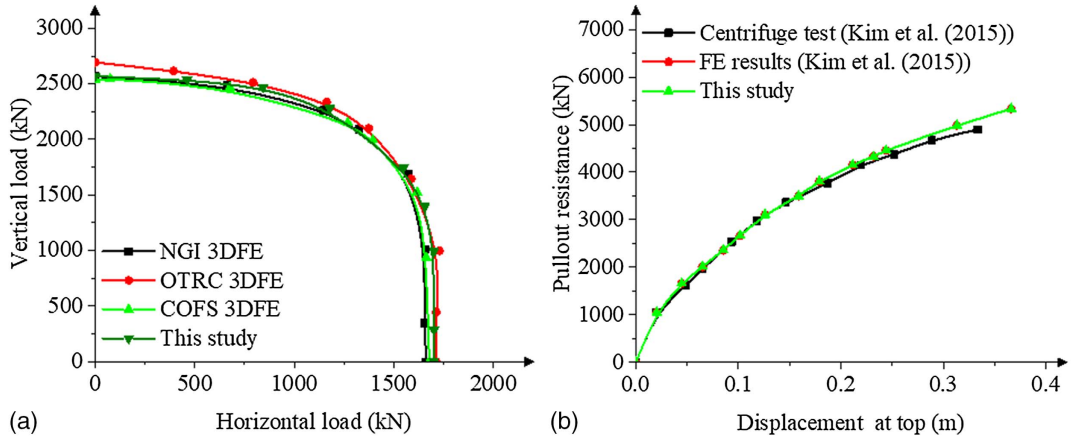

Fig. 5. Model validation against previous studies: (a) pure clay ($L/D = 1.5$, $s_u = 1.25z$); and (b) pure silty sand ($L/D = 2$, $c_u = 7$ kPa, $\phi = 32°$, $\psi = 8°$).

predictions were performed by several reputational organizations including the Norwegian Geotechnical Institute (NGI), Offshore Technology Research Center (OTRC), and Centre for Offshore Foundation Systems (COFS). In this case, $s_u = 1.25z$ and the anchor has a length ratio of $L/D = 1.5$. Fig. 5(a) compares the VH yield envelope modeled in this study with the others' predictions. It can be found that the results agree reasonably, which demonstrates the numerical modeling method in this study could well replicate the behavior of suction anchors in NC clay.

For suction anchors in silty sand, previous centrifuge testing data and FE modeling results by Kim et al. (2015) were employed for comparison. The anchor was subjected to inclined loading at its padeye ($Z_{pad} = 2/3L$), soil properties were selected as $c_u = 7$ kPa, $\phi = 32°$, and $\psi = 8°$, and the anchor aspect ratio was $L/D = 2$. Fig. 5(b) compares the displacements of the anchor, where the results from this study are close to those of Kim et al. The anchor response in the FE modeling is slightly higher than that in the centrifuge test. The main reason may be that the modelling of the mooring chain is not involved, which nevertheless has no effect on the establishment of the VH yield envelopes.

Overall, the preceding comparisons demonstrate the numerical method used in this study can provide confidence for predictions of foundations in clay and silty sand under inclined loading conditions.

Failure Mechanism

To better illustrate the failure mechanism of suction anchors in a clay-over-silty-sand deposit, reference cases with a suction anchor installed in NC clay and silty sand are included for comparison ($s_{um}/kD = 0$, $c_u/\gamma' D = 0.34$, $L/D = 4.5$, and $T_c/D = 2$). Fig. 6(a) shows the failure mechanisms of the suction anchor in pure NC clay, pure silty sand, and clay over silty sand. Meanwhile, to clearly elaborate the failure mechanism from previous studies, one typical stratified soil profile (clay over sand) was also involved for comparison. The layered sand is classified as silica sand obtained through numerous investigations from University of Western Australia (UWA). Detailed sand layer properties can be found in Zou et al. (2018). The anchor is under pure horizontal translation without rotation. From the displacement vector, it can be seen that the failure mechanisms of the anchor in pure NC clay, silty sand, and clay over silty sand are significantly different. For the pure NC clay, the soil flow on the back side of the anchor is still mobilized even if the gap interface is considered. For the pure silty sand, due to the existence of the cohesion, the soil can maintain its stability and gapping can occur between the soil and anchor. Accordingly, in the case of the anchor in clay over silty sand, the soil flow shows a clear boundary between the clay and the silty sand. On the rear side of the anchor when it is in clay over silty sand, the tension gap is developed in the silty sand layer while the remainder is enclosed in clay. This is because for NC clay with low strength at the seabed, the soil behind the anchor will collapse by the anchor movement, and then following the movement of the anchor. In comparison, when the anchor was subjected to clay over sand, the gapping condition will stay closed because silica sand can collapse with the movement of the anchor.

With different soil failure mechanisms, the reaction forces along the anchor differ as expected. Fig. 6(b) shows the unit reaction force along the anchor in clay, silty sand, and clay over silty sand, respectively. Overall, the reaction force increases along the anchor elevation from zero at the seabed to the maximum values at the tip of the anchor. As shown in Fig. 6(b), when the anchor is in the pure NC clay or in the overlying clay of clay over silty sand, the soil reaction force shows similar trends and increases slowly with the increase of depth. However, when the anchor is in clay over silty sand, the reaction force along the anchor shows a kink at the clay–silty sand interface. With the increasing anchor length in silty sand, the reaction force of the anchor in clay over silty sand is increasing dramatically and shows a similar tendency to the reaction force in pure silty sand. This trend shows that the existence of the silty sand layer will significantly influence the reaction force of the anchor. Therefore, a higher pullout capacity can be obtained when the anchor is in the silty sand layer.

Results and Discussion

A parametric study was performed and the FE results are given in this section. The optimum centerline loading depth $Z_{cl,opt}$ of suction anchors in clay over silty sand is identified, which is followed by the ultimate uniaxial pullout capacities under horizontal and vertical loading. Then the distribution of the pullout capacity under the horizontal loading at different padeye locations is presented, and the VH failure envelopes of suction anchors in clay over silty sand under inclined loading are constructed.

Optimum Centerline Loading Depth $Z_{cl,opt}$

The optimum centerline loading depth $Z_{cl,opt}$ is a critical factor determining the translational and rotational movements of an anchor.

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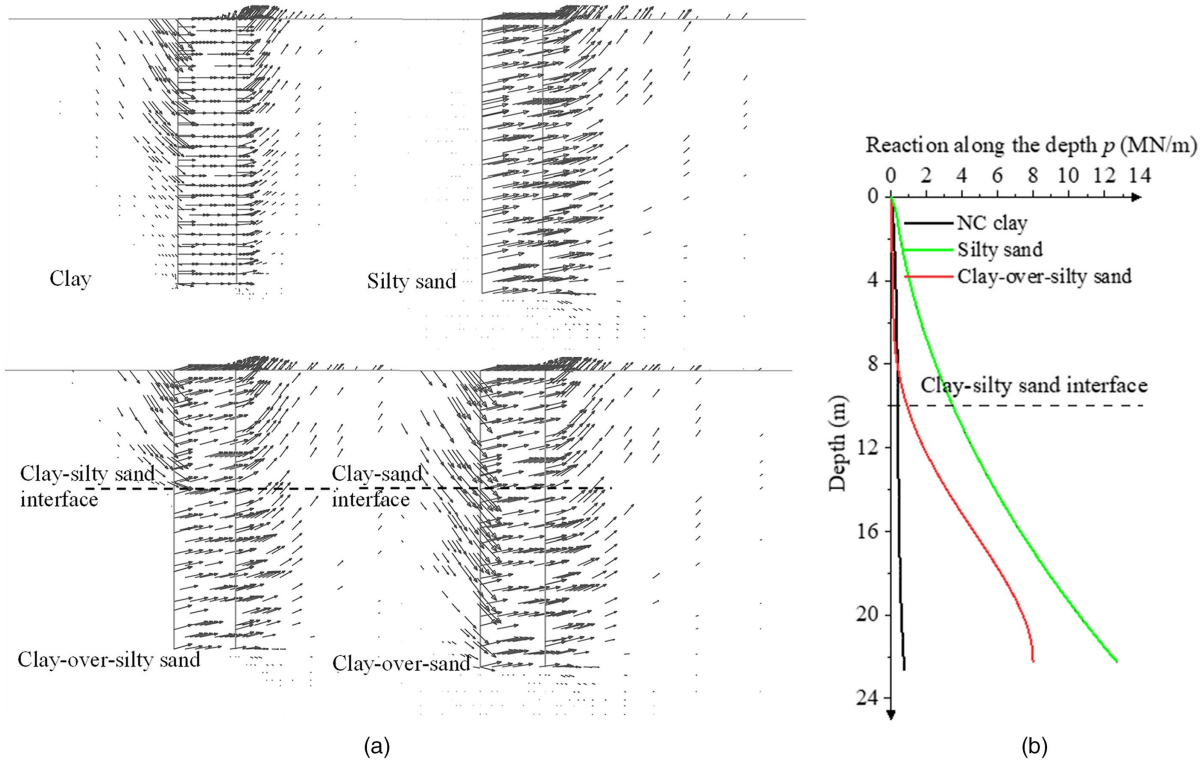

Fig. 6. Failure mechanisms and reaction force of anchors ($s_{um}/kD = 0$, $c_u/\gamma' D = 0.34$, $L/D = 4.5$, and $T_c/D = 2$): (a) soil failure mechanisms; and (b) reaction force.

When the inclined loading vector goes through $Z_{cl,opt}$, the anchor can translate horizontally and reach its maximum pullout capacity. When the inclined loading lies above or below the optimum centerline loading depth $Z_{cl,opt}$, the anchor will rotate forward or backward accordingly. Thus, the optimum centerline loading depth $Z_{cl,opt}$ can be used as an indication to reflect the failure mode and the maximum pullout capacity, and it is necessary to get the $Z_{cl,opt}$ position in the design process.

Fig. 7 shows several typical normalized optimum centerline loading depths $Z_{cl,opt}/L$, in which the soil property varies as $s_{um}/kD = 0$ and 1, $c_u/\gamma' D = 0.07$ and 0.34, and the relative position of the anchor varies as $L/D = 1.5$–6, $T_c/D = 3.0$. It is apparent that soil properties of the silty sand, $c_u/\gamma' D$, have negligible effect on $Z_{cl,opt}/L$. Thus, in the following investigation, only $c_u/\gamma' D = 0.34$ is considered. Fig. 8 shows the normalized optimal

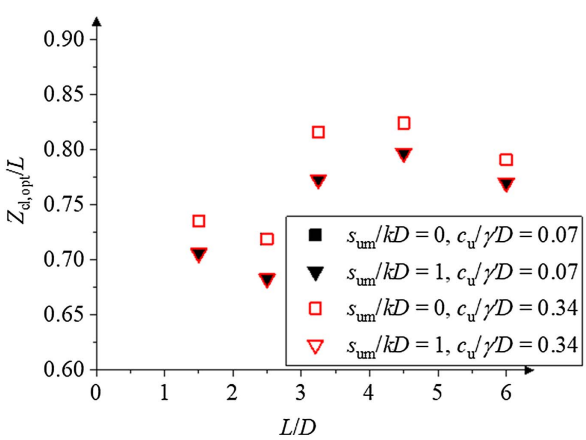

Fig. 7. Effect of soil properties on $Z_{cl,opt}$.

centerline loading depth $Z_{cl,opt}/L$ of the suction anchor in clay over silty sand. Figs. 8(a–c) present the effect of the clay thickness ratios T_c/D of 1.0, 2.0, and 3.0 on $Z_{cl,opt}/L$, respectively. The vertical dashed line denotes the interface between the clay and the silty sand, in which $Z_{cl,opt}/L$ of the suction anchor in the overlying clay are on the left and $Z_{cl,opt}/L$ of suction anchors in clay over silty sand are on the right. Overall, the normalized optimum centerline loading depth $Z_{cl,opt}/L$ is only related to the overlying clay properties and the relative position of the suction anchor in clay over silty sand.

For the overlying clay properties, with the increase of clay strength s_{um}/kD, $Z_{cl,opt}/L$ will decrease accordingly. This is due to the increase in clay strength s_{um}/kD, leading to the increase of the reaction force of the anchor in clay to the overall reaction force, which thus results in $Z_{cl,opt}/L$ to move up. For the relative position of the suction anchor in clay over silty sand, Numerical modelling can be analyzed in two conditions: the tip of the anchor installed in the overlying clay or in the silty sand layer. When the suction anchor is located in the overlying clay, the $Z_{cl,opt}/L$ ranges from 0.66 to 0.74. Similar conclusions can be found for suction anchors in pure clay in previous studies (Supachawarote 2006; Fu et al. 2020). When the tip of the anchor lays in the silty sand layer, the $Z_{cl,opt}/L$ ranges from 0.75 to 0.85. This can be contributed to the fact that the silty sand provides a larger resistance to the suction anchor leading to the change of $Z_{cl,opt}$. Thus, it is obvious that with the existence of the silty sand layer, $Z_{cl,opt}/L$ will increase, forming a demarcation on $Z_{cl,opt}$ between the clay and the silty sand, as shown in Fig. 8.

Furthermore, based on the correlation of $Z_{cl,opt}/L$ and the relative position of the suction anchor in the clay over silty sand, there will be a transition zone on $Z_{cl,opt}/L$. When the tip of the anchor is located in the clay or only a small part of the anchor is in the silty sand, the clay properties play a dominant role on $Z_{cl,opt}/L$. With the

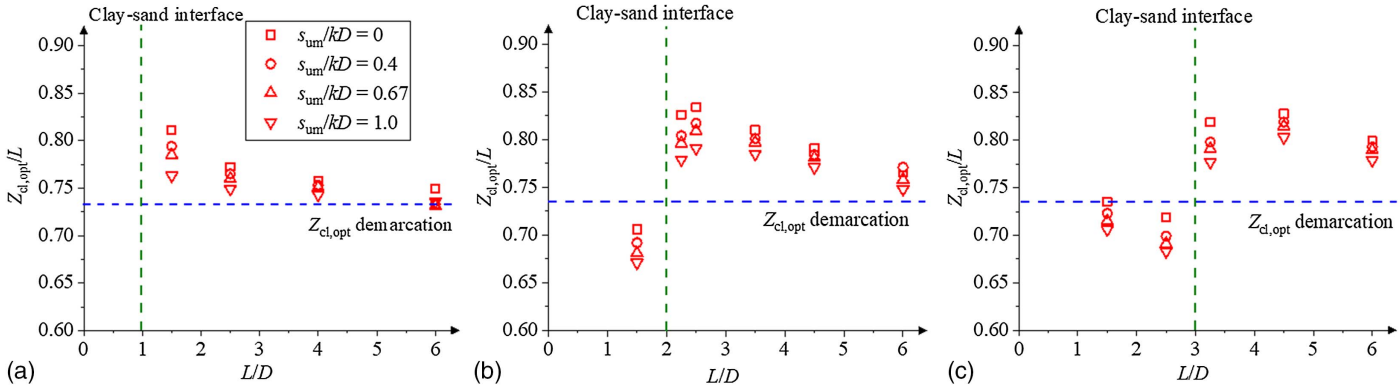

Fig. 8. Optimum centerline loading depth $Z_{cl,opt}$: (a) $T_c/D = 1.0$; (b) $T_c/D = 2.0$; and (c) $T_c/D = 3.0$.

increasing length of anchor in the silty sand, the effect of clay properties on $Z_{cl,opt}/L$ diminishes. As shown in Fig. 8(b), when the length of the anchor is relatively short ($L/D < 3$, $T_c/D = 2.0$), the effect of clay properties s_{um}/kD on $Z_{cl,opt}/L$ is obvious. When the anchor tip enters deeply into the silty sand, the influence of clay properties s_{um}/kD on $Z_{cl,opt}/L$ can be ignored.

Ultimate Uniaxial Pullout Capacity (H_{ult} and V_{ult})

The ultimate uniaxial pullout capacity (H_{ult} and V_{ult}) can be treated as special cases of the anchors under the horizontal or vertical movement without rotation. In the simulation, this is equivalent to the integration of the limiting lateral resistance of the soil along the depth of the anchor (Randolph and House 2002; Aubeny et al. 2003). Following this principle, the relative position of the anchor (L/D and T_c/D) and the soil properties (s_{um}/kD and $c_u/\gamma' D$) on the ultimate uniaxial pullout capacities (H_{ult} and V_{ult}) are evaluated, and the FE results are plotted in Figs. 9 and 10, respectively.

Fig. 9 shows the ultimate horizontal pullout capacity factors N_{ch} varying with the relative position of the anchor and the soil properties according to the overlying clay thickness ratio T_c/D. The horizontal dashed line denotes the clay–silty sand interface. For simplicity, Fig. 9 only shows partial results of the silty sand properties ($c_u/\gamma' D = 0.07$ and 0.34).

In general, it can be found that N_{ch} increases with the increasing L/D ratio and the rise of the soil strength (s_{um}/kD and $c_u/\gamma' D$), indicating a longer anchor and a stronger soil can improve the load-resistance capacity of the anchor. However, with the existence of the silty sand, the relative position of the anchor in clay over silty sand can be varied, and the effect on N_{ch} can be slightly different. As shown in Fig. 9, when the tip of the anchor is located above the clay–silty sand interface, N_{ch} gradually increases with the rise of L/D and s_{um}/kD. This suggests that N_{ch} is only related to the length of the anchor L/D and clay properties s_{um}/kD, but not silty sand properties $c_u/\gamma' D$. When the tip of the anchor is beneath the clay–silty sand interface, N_{ch} can increase rapidly. This demonstrates that with the existence of the silty sand, the ultimate horizontal pullout capacities of suction anchors will be greatly improved. This is because the transformation of the soil flow mechanism of the low-shear-strength clay is shifted to a deeper soil flow mechanism of the silty sand as shown in Fig. 6.

Furthermore, with the increase of the clay thickness ratio T_c/D, N_{ch} decreases accordingly, and the effects of soil properties on N_{ch} can reduce. For example, as shown in Fig. 9(a) when $T_c/D = 1.0$ and Fig. 9(c) when $T_c/D = 3.0$, the differences between $c_u/\gamma' D = 0.07$ and 0.34 on N_{ch} are decreasing appreciably with the increase of T_c/D. Such change in the ultimate horizontal pullout capacity of suction anchors is mainly because of the decreasing

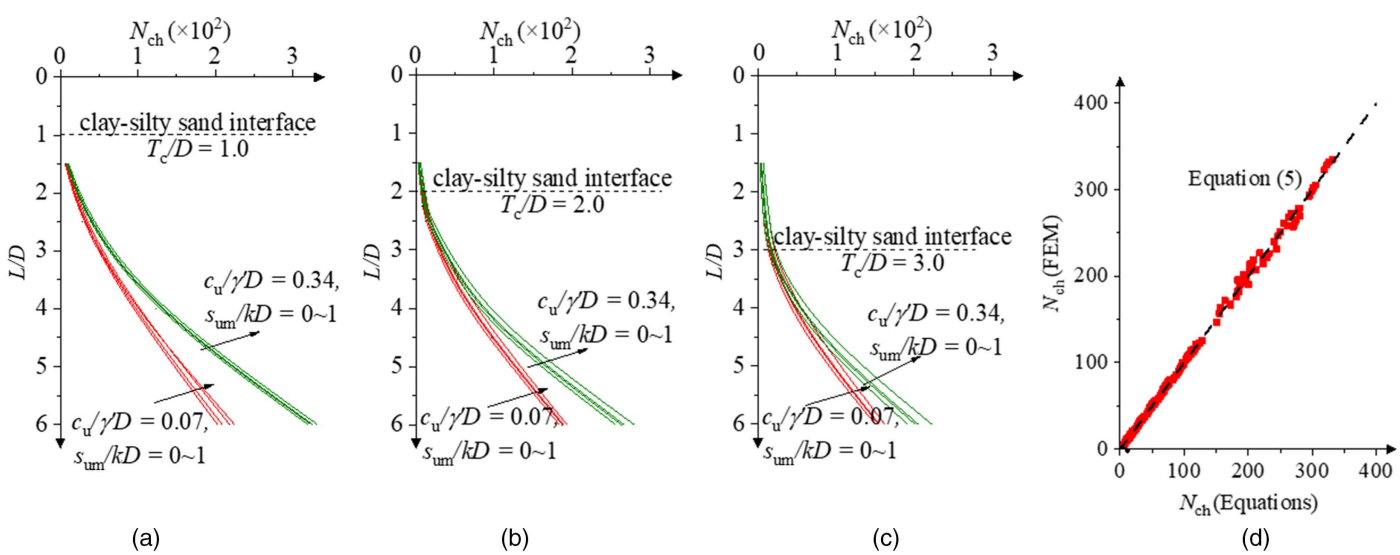

Fig. 9. Horizontal pullout capacity: (a) $T_c/D = 1.0$; (b) $T_c/D = 2.0$; (c) $T_c/D = 3.0$; and (d) fitting degree.

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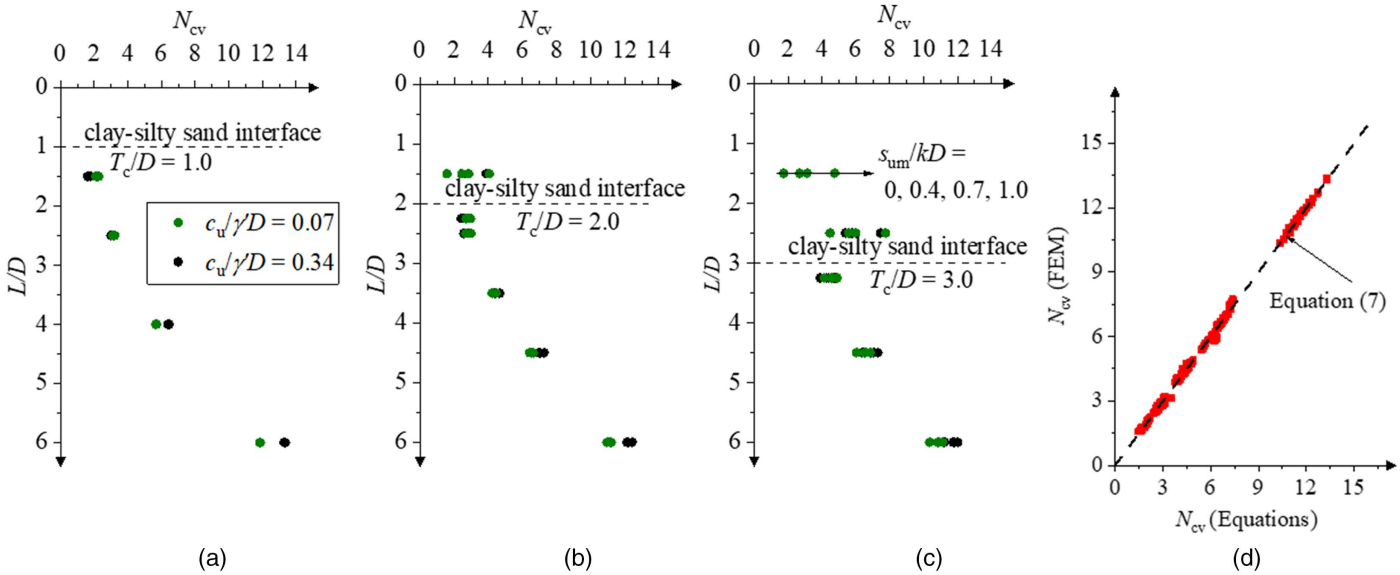

Fig. 10. Vertical pullout capacity: (a) $T_c/D = 1.0$; (b) $T_c/D = 2.0$; (c) $T_c/D = 3.0$; and (d) fitting degree.

proportion of the anchor length in the silty sand with the increase of T_c/D.

For the ultimate vertical pullout capacity, Fig. 10 shows N_{cv} can also be mobilized by longer anchors and increasing soil strength. Various T_c/D ratios lead to different lateral friction distributions of clay and silty sand, which influence the ultimate vertical pullout capacities of the anchor. For example, in the cases of $T_c/D = 2.0$ as in Fig. 10(b) and $T_c/D = 3.0$ in Fig. 10(c), the proportion of the anchor in the silty sand will decrease with increase of T_c/D, and the change on N_{cv} will be formed. Thus, the location of the anchor tip in clay over silty sand clearly significantly influences N_{cv}.

When the tip of the anchor is above the clay–silty sand interface, being similar to N_{ch}, N_{cv} can increase slowly with the increasing L/D and s_{um}/kD. The silty sand properties $c_u/\gamma'D$ have negligible influence on N_{cv}. When the tip of the anchor is in the silty sand layer below the clay–silty sand interface, the ultimate vertical pullout capacity consists of two components: anchor–clay soil resistance and anchor–silty sand soil resistance. The clay properties s_{um}/kD and sand properties $c_u/\gamma'D$ both have obvious influence on N_{cv}. For example, when the clay thickness ratio T_c/D is less than 1.0 as shown in Fig. 10(a), the lateral resistance of the anchor in the silty sand contributes a dominant proportion of the resistance on N_{cv}. When the clay thickness T_c/D is more than 2.0 as shown in Figs. 10(b and c), the influence of the anchor–clay soil resistance cannot be ignored.

Based on the preceding simulation results, the ultimate uniaxial pullout capacity factors (N_{ch}, N_{cv}) can be defined with consideration of the anchor length ratio (L/D), the overlying clay thickness ratio (T_c/D), and soil properties ($c_u/\gamma'D$ and s_{um}/kD). Eqs. (4)–(7) are introduced to describe N_{ch} and N_{cv} for the anchors in clay over silty sand. Verification is performed, where it can be seen from Figs. 9(d) and 10(d) that the predictions using these equations are falling well within the numerical modeling results. The following expressions are derived by fitting the FE results from the parametric study:

$$H_0 = N_{ch}\gamma' D^3 \tag{4}$$

$$N_{ch} = a + b \times \left(\frac{s_{um}}{kD}\right) + c \times \log\left(\frac{c_u}{\gamma' D}\right) \tag{5}$$

Table 4. Coefficient of ultimate uniaxial horizontal pullout capacity

T_c/D	L/D	a	b	c
1	1.5	8.74	2.43	1.20
1	2.5	36.05	4.71	4.48
1	4	124.75	7.62	15.16
1	6	369.41	12.12	68.24
2	1.5	2.27	3.64	0.00
2	2.25	9.41	4.40	0.69
2	2.5	15.30	5.59	1.38
2	3.5	52.56	10.66	5.11
2	4.5	118.72	15.22	12.98
2	6	294.09	17.92	48.77
3	1.5	2.47	4.16	0.00
3	2.5	6.66	7.30	0.00
3	3.25	16.07	7.76	0.81
3	4.5	72.27	17.25	6.09
3	6	217.10	23.86	30.69

$$V_0 = N_{cv}\gamma' D^3 \tag{6}$$

$$N_{cv} = a + b \times \left(\frac{s_{um}}{kD}\right) + c \times \log\left(\frac{c_u}{\gamma' D}\right) \tag{7}$$

where the fitting coefficients are given in Tables 4 and 5.

Horizontal Pullout Capacity Distribution at Different Padeyes

Because different padeye positions can lead to anchor rotation, the horizontal pullout capacity at the padeye H_0 will be different from the ultimate horizontal capacity of the anchor H_{ult}. To determine H_0 of the anchor in clay over silty sand, a group of analyses were conducted with $Z_{pad}/L = 0.3$–0.9. Additionally, because H_{ult} can be predicted using Eqs. (4) and (5), the distribution of the horizontal capacity at the padeye can be expressed using H_0/H_{ult}.

Fig. 11 presents the variations in the normalized H_0/H_{ult} distributions with various relative positions of the anchor (T_c/D and L/D) and soil properties ($c_u/\gamma'D$ and s_{um}/kD). Overall, the normalized H_0/H_{ult} distribution follows a skewness curve. With varying ratio of Z_{pad}/L, H_0/H_{ult} can reach the maximum value

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Table 5. Coefficient of ultimate uniaxial vertical pullout capacity

T_c/D	L/D	a	b	c
1	2.37	2.37	0.17	0.31
1	3.19	3.19	−0.04	0.05
1	5.16	5.16	0.01	−0.46
1	10.78	10.78	0.02	−0.95
2	1.59	1.59	2.30	0.04
2	2.84	2.84	0.22	0.17
2	2.65	2.65	0.28	0.05
2	4.05	4.05	0.22	−0.11
2	6.01	6.01	0.24	−0.35
2	10.25	10.25	0.23	−0.70
3	1.57	1.57	2.90	0.00
3	4.52	4.52	2.97	0.11
3	4.27	4.27	0.78	0.13
3	5.74	5.74	0.83	−0.26
3	9.82	9.82	0.72	−0.54

1.0 at the optimum padeye, which then gradually decreases. This is because when the applied loading is above or below the optimum padeye, the rotation of the anchor will reduce the horizontal capacity. Moreover, the proportion of the anchor length in the silty sand also has an obvious effect on the normalized H_0/H_{ult} distributions. For example, as shown in Fig. 11(a), when the anchor tip is in clay (e.g., $T_c/D = 3.0 > L/D = 2.5$) or the proportion of the anchor length in the silty sand is less than $0.5D$ (e.g., $T_c/D = 1.5 < L/D = 2.0$), the lateral resistance of the anchor in clay plays a dominant role, and the normalized H_0/H_{ult} distribution can be expressed by a unique skewness curve. When the anchor tip is in the silty sand and the proportion of the anchor length in the silty sand is more than $0.5D$ (e.g., $T_c/D = 2.0 < L/D = 3.5$) as shown in Fig. 11(b), the lateral resistance of the anchor in the silty sand is dominant with another type of unique skewness curves. Therefore, it can be seen that the main difference between Figs. 11(a and b) is the proportion of the horizontal resistance between the clay and the silty sand, which thus influences the unique trends of the normalized H_0/H_{ult} distribution.

The next step is to quantify the characteristics of the unique curves of the normalized H_0/H_{ult} distribution in clay over silty sand. As shown in Figs. 11(a and b), even with the changing optimum padeye positions, the decreasing trend of the unique skewness curves remains roughly constant. Therefore, the dashed lines can be fitted and kept constant as the $Z_{pad,opt}$ changes. Besides, the reduction degree of the H_0/H_{ult} distribution in Fig. 11(a) is smaller than that in Fig. 11(b). This can be owing to the lower shear strength of clay: when the suction anchor moves forward or backward, the anchor displacement mobilizes more soil to provide resistance. Based on the preceding results, the normalized H_0/H_{ult} distribution in the clay over silty sand can be described by two unique skewness curves. When the tip of the anchor is in clay or the proportion of the anchor length in the silty sand is less than $0.5D$, the normalized H_0/H_{ult} distribution can be obtained using the following equation:

$$\frac{H_0}{H_{ult}} = \begin{cases} e^{0.016+1.91\times(Z_{pad}-Z_{pad,opt})} & 0.3 < Z_{pad} \le Z_{pad,opt} \\ \ln(2.72 - 3.45 \times (Z_{pad} - Z_{pad,opt})) & Z_{pad,opt} < Z_{pad} \le 0.9 \end{cases} \tag{8}$$

When the tip of the anchor is in the silty sand and the proportion of the anchor length in the silty sand is more than $0.5D$, the normalized H_0/H_{ult} distribution can be obtained using the following equation:

$$\frac{H_0}{H_{ult}} = \begin{cases} e^{-0.03+2.72\times(Z_{pad}-Z_{pad,opt})+0.53\times(Z_{pad}-Z_{pad,opt})^2} & 0.3 < Z_{pad} \le Z_{pad,opt} \\ \ln(2.69 - 3.78 \times (Z_{pad} - Z_{pad,opt})) & Z_{pad,opt} < Z_{pad} \le 0.9 \end{cases} \tag{9}$$

where Z_{pad} = loading position ranging from 0.3 to $0.9L$; and $Z_{pad,opt}$ = optimum padeye position, which can be found in the "Optimum Centerline Loading Depth $Z_{cl,opt}$" section. The individual influencing factors (T_c/D, L/D, $c_u/\gamma' D$, and s_{um}/kD) are not shown in these equations because their influences have been included in the process of determining $Z_{pad,opt}$.

VH Failure Envelopes

After obtaining the ultimate uniaxial pullout capacities (H_{ult} and V_{ult}) and the normalized H_0/H_{ult} distribution, the following step can be followed to build the VH failure envelopes. However, because of different failure mechanisms and bearing behaviors of the anchors in clay-over-silty-sand layered deposits, the VH failure

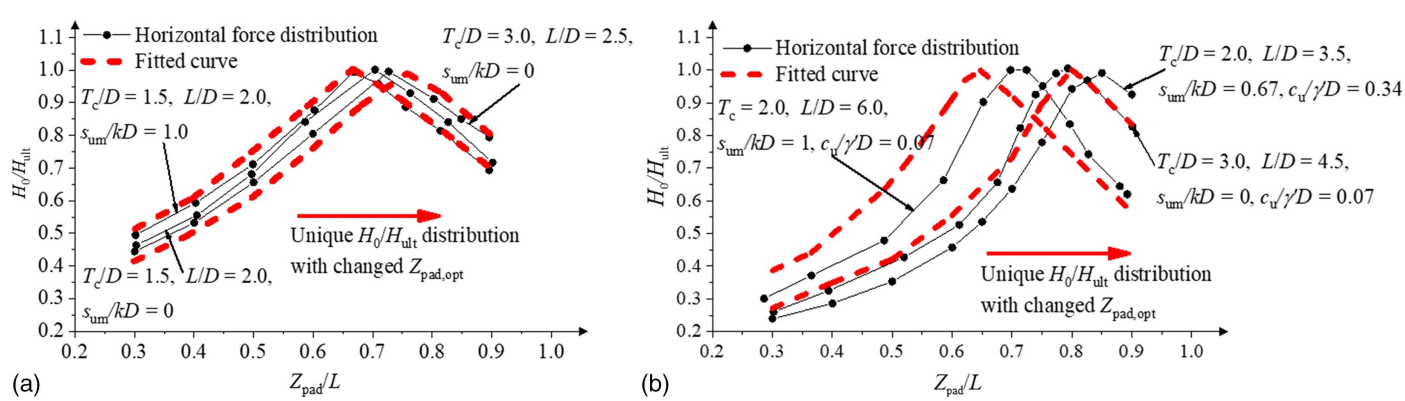

Fig. 11. Normalized H_0/H_{ult} distributions: (a) H_0/H_{ult} for the clay lateral resistance-dominant cases; and (b) H_0/H_{ult} for the silty sand lateral resistance-dominant cases.

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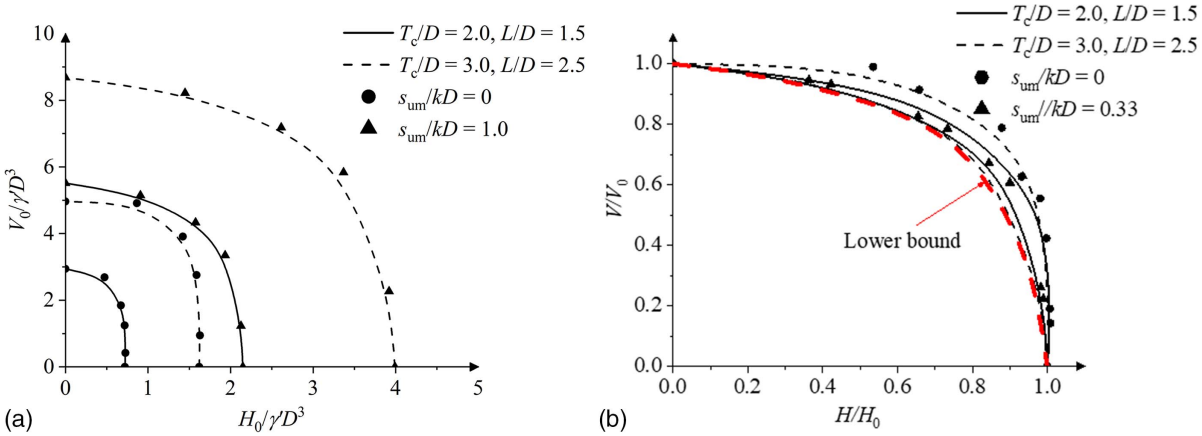

Fig. 12. *VH* failure envelopes when anchor tip is in clay: (a) *VH* failure envelopes; and (b) normalized *VH* failure envelopes.

envelopes can also differ in size and shape. Thus, further numerical modeling was conducted in two categories: (1) *VH* failure envelopes for the tip of the anchor in the overlying clay; and (2) *VH* failure envelopes for the tip of the anchor in the silty sand. Group 6 (Table 1) shows all the selected cases with varying suction anchor length L/D, clay thickness T_c/D, and soil properties (s_{um}/kD and $c_u/\gamma' D$).

For the cases with the anchor tip in clay, it is common practice in practical designs to restrict the rotation of the anchor in determining the ultimate limit state (Randolph et al. 1998; Supachawarote 2006). The benefit of this approach is that the ultimate *VH* failure envelopes can be identified in a single analysis, which is particularly suitable for short and rigid anchors. Therefore, the anchors with tips in the overlying clay are restrained to rotate. Fig. 12(a) shows typical failure envelopes, in which *VH* envelopes expand in size with the increase of L/D and s_{um}/kD. Meanwhile, to visually show the relative shapes of the *VH* envelopes, the results are also normalized by the uniaxial loads (H_0 and V_0). Apparently, a unique lower bound can be observed, suggesting that the normalized *VH* failure envelopes are non-associated with anchor length and soil properties. This conservative trend is very useful because the yield envelopes in the *VH* domain can be simply obtained by scaling up H_0 and V_0. Similar trends were also observed in previous studies of the bearing behaviors for spudcans and bucket foundations (Gourvenec and Randolph 2003; Gourvenec 2008; Zhang et al. 2011, 2012; Hung and Kim 2014). Based on the preceding results and some regression analyses, the lower bound of *VH* can be expressed as

$$\frac{H_0}{H_{ult}} = \left\{1 - \left(\frac{V_0}{V_{ult}}\right)^3\right\}^{1/1.38} \tag{10}$$

where V_0/V_{ult} = normalized vertical pullout capacity; and H_0/H_{ult} = normalized horizontal pullout capacity. The exponent of 1/3.38 demonstrates the vertical load would reduce the horizontal loading capacity.

For the cases with anchor tip in the silty sand, Figs. 13(a–d) show the yield envelopes with changing padeye depth Z_{pad}, the relative position of the anchor (T_c/D and L/D), and soil properties s_{um}/kD and $c_u/\gamma' D$, respectively. As shown, due to different degrees of *VH* interaction in the silty sand, the *VH* space envelopes cannot be expressed with one unique lower-bound expression as previously. Instead, the protruding envelopes as shown in Fig. 13 can be roughly determined by three factors: the uniaxial pullout capacity (H_0 and V_0), the cone top of the envelopes denoted as V_{max}, and the loading angle θ_{max} where V_{max} occurs. Therefore, to determine the *VH* failure envelopes in the silty sand, the next step is to identify these three influencing factors.

Eqs. (4)–(9) can be used to obtain the uniaxial pullout capacities (H_0 and V_0). First, the ultimate uniaxial pullout capacities (H_{ult} and V_{ult}) can be obtained using Eqs. (4)–(7). Then, according to the normalized H_0/H_{ult} distribution, the uniaxial pullout capacities (H_0 and V_0) can be obtained at any given padeye positions using Eqs. (8) and (9). As shown in Fig. 13, V_{max} appears nearly at half of the horizontal pullout capacity H_0, regardless of the relative position of the anchor and soil properties. Previous studies also showed similar trends when anchors were in silica sand and silty sand (Zhao et al. 2019; Cheng et al. 2021). In the cases of the anchor tip in the silty sand, the relative shape of the *VH* envelopes is only influenced by the loading angle θ_{max}. Moreover, as shown in Figs. 13(c and d), the soil properties (s_{um}/kD and $c_u/\gamma' D$) only influence the absolute *VH* envelopes, whereas there is no effect on loading angle θ_{max}. Therefore, the solution for θ_{max} only depends on the loading position Z_{pad}/L and the relative position of the anchor (T_c/D and L/D). Table 6 lists the numerical modeled loading angle θ_{max} through a series of FE analysis.

Proposed Design Procedure

To assess the pullout capacity of suction anchors in clay-over-silty-sand deposits, a design framework is proposed based on the numerical analysis in this study. Fig. 14 gives the suggested design procedure, which can be outlined as follows:

1. From the geological reports and installation data, obtain the anchor length ratio L/D and the overlying clay thickness ratio T_c/D; and then determine the soil properties of the clay-over-silty-sand deposit (s_{um}/kD and $c_u/\gamma' D$) from geological data [standard penetration test (SPT) or cone penetration test (CPT) data].
2. Calculate the optimal centerline loading depth $Z_{pad,opt}$ referring to Fig. 8; and obtain the optimum padeye depth $Z_{pad,opt}$ using Eq. (3).
3. Determine the ultimate uniaxial pullout capacities V_{ult} and H_{ult} referring to Eqs. (4)–(7) with the corresponding coefficients.
4. Calculate the horizontal pullout capacity H_0 at the given padeye depth Z_{pad} using Eqs. (8) and (9).
5. Determine the relative position of the suction anchor; then obtain the relative shape of the *VH* envelopes. When the tip of the anchor is in the overlying clay, construct the normalized *VH* failure surfaces with the lower bound using Eq. (10). When the tip of the anchor is in the silty sand, the loading angle θ_{max} is introduced to depict the relative shape of the *VH* failure surface.

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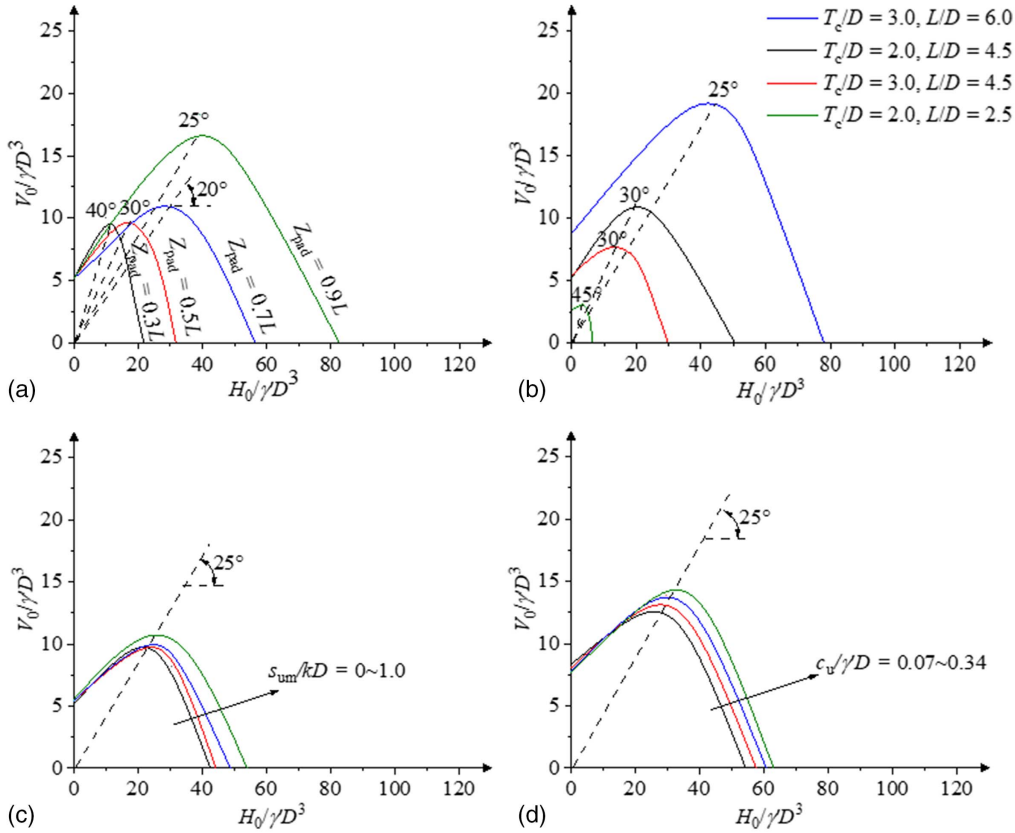

Fig. 13. VH failure envelopes for anchor tip in silty sand: (a) VH envelopes with various Z_{pad} ($T_c/D = 3$, $L/D = 6$, $s_{um}/kD = 0$, $c_u/\gamma' D = 0.34$); (b) VH envelopes with various T_c/D and L/D ($s_{um}/kD = 0$, $c_u/\gamma' D = 0.34$, $Z_{pad}/L = 0.6$); (c) VH envelopes with various s_{um}/kD ($T_c/D = 3$, $L/D = 6$, $c_u/\gamma' D = 0.34$, $Z_{pad}/L = 0.6$); and (d) VH envelopes with various $c_u/\gamma' D$ ($T_c/D = 3$, $L/D = 6$, $s_{um}/kD = 0$, $Z_{pad}/L = 0.6$).

Table 6. Maximum loading angle θ_{max}

		θ_{max}						
T_c/D	L/D	$Z_{pad}/L = 0.3$	$Z_{pad}/L = 0.4$	$Z_{pad}/L = 0.5$	$Z_{pad}/L = 0.6$	$Z_{pad}/L = 0.7$	$Z_{pad}/L = 0.8$	$Z_{pad}/L = 0.9$
1	1.5	50	45	40	45	45	30	30
1	2.5	35	35	30	25	25	25	30
1	4	30	35	20	20	20	20	30
1	6	30	30	25	20	20	20	30
2	2.25	60	60	55	50	40	35	35
2	2.5	60	50	50	45	40	30	30
2	3.5	40	35	40	25	25	25	25
2	4.5	35	30	30	30	23	20	25
2	6	35	30	30	25	20	20	25
3	3.25	65	60	50	45	40	30	45
3	4.5	45	45	35	30	23	20	25
3	6	40	35	30	25	20	20	30

6. Construct the absolute VH failure envelopes by utilizing the uniaxial pullout capacities.

To evaluate the accuracy of the proposed approach, the combined failure surfaces were conducted and compared with the numerical modeling. Fig. 15 depicts one typical VH failure surface for the cases of $L/D = 4.5$, $T_c/D = 2.0$, $Z_{pad}/L = 0.3$–0.9, $s_{um}/kD = 0$, and $c_u/\gamma' D = 0.13$. A minor discrepancy can be found on the FE results and the proposed method, which demonstrates the proposed method could properly and reasonably generate the VH envelopes of suction anchors in clay-over-silty-sand deposits.

Conclusions

This study presented intensive numerical modeling to investigate the pullout bearing behaviors of suction anchors in clay over silty sand under inclined loading. Numerical modeling was carried out to determine the optimum padeye position and the uniaxial pullout capacities of suction anchors. The combined VH failure envelopes of the anchors in clay over silty sand were established. A simplified design procedure was proposed to calculate the inclined pullout capacities of suction anchors in clay-over-silty-sand deposits. This study investigated the behavior of anchors in clay overlying silty

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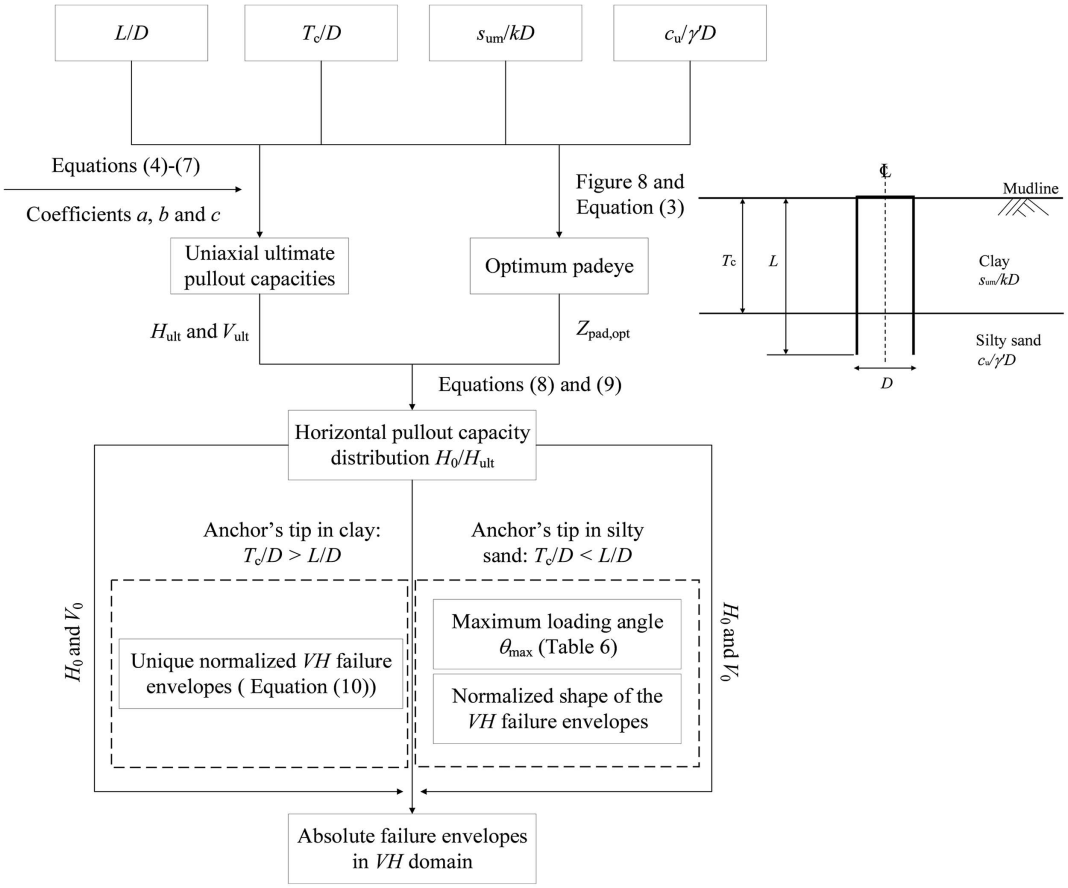

Fig. 14. Proposed design procedure.

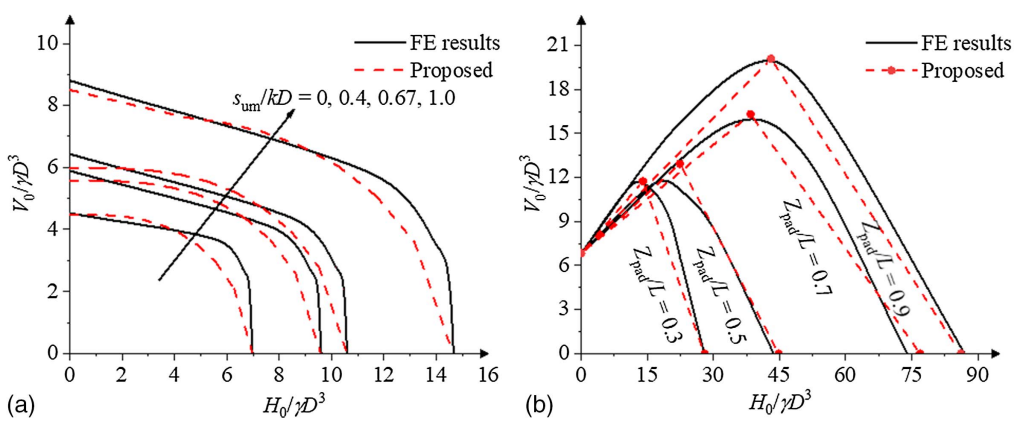

Fig. 15. Comparison of VH envelopes between the proposed equations and FE analyses: (a) VH envelopes for anchor tip in clay ($L/D = 2.5$ and $T_c/D = 3$); and (b) VH envelopes for anchor tip in silty sand ($s_{um}/kD = 0.4$, $c_u/\gamma' D = 0.13$, $L/D = 4.5$, and $T_c/D = 2$).

sand, which gives a design guidance for predicting the pullout capacity of anchors in engineering applications. The following are detailed conclusions:

- The failure mechanisms and the ultimate pullout capacities of suction anchors in clay over silty sand differ significantly from those in pure clay. The existence of silty sand can restrict the soil flow, which improves the bearing capacity of the suction anchor.
- The optimum padeye depth of the suction anchor in clay over silty sand mostly depends on the relative position of the anchor. For $T_c < L$, when the tip of the anchor is in the overlying clay, the normalized optimum padeye position $Z_{pad,opt}/L$ varies from ranges from 0.66 to 0.74. For $T_c > L$, when the tip of the anchor is in the silty sand, the normalized optimum padeye position $Z_{pad,opt}/L$ ranges from 0.75 to 0.85.
- In general, the uniaxial pullout capacities (H_{ult} and V_{ult}) are found to increase with the anchor length and the soil strength. The proportion of the anchor length in the silty sand also has an obvious effect on the pullout capacity. When the tip of the

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anchor transitions from clay to silty sand, the pullout capacities can be improved sharply. Eqs. (4)–(7) can be used to estimate the uniaxial pullout capacities of suction anchors (V_0 and H_0) in clay-over-silty-sand deposits.

- The normalized H_0/H_{ult} distribution of the anchors in clay over silty sand can be expressed as unique skewness curves, regardless of soil properties. And the shape of the unique skewness curves depends on the horizontal resistance proportion on the clay and the silty sand.
- The failure surfaces and the normalized envelopes in the VH domain are constructed to estimate the ultimate pullout abilities of the anchors in clay-over-silty-sand deposits under inclined loading. The proposed design procedure can be used to guide design of suction anchors in clay-over-silty-sand deposits.

Data Availability Statement

The detailed data sets and models generated during the current study are available from the corresponding author upon reasonable request.

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Notation

The following symbols are used in this paper:

a, b, c = fitting parameters;
B_H = distance of horizontal boundary;
B_V = distance of vertical boundary;
c_u = undrained shear strength of silty sand;
$c_u/\gamma' D$ = normalized shear strength of silty sand;
D = diameter of foundation;
E = Young's modulus;
H = horizontal load;
H_{ult} = ultimate horizontal pullout capacity;
H_0 = horizontal pullout capacity;
H_0/H_{ult} = normalized horizontal pullout capacity;
$H_0/\gamma' D^3$ = dimensionless horizontal pullout capacity;
k = strength gradient of clay;
L = length of suction anchor;
L/D = length ratio of suction anchor;
M = reaction moment;
N_{ch} = horizontal pullout capacity factor;
N_{cv} = vertical pullout capacity factor;
p = reaction force along the depth;
s_u = undrained shear strength of clay;
s_{um} = undrained shear strength at mud line;
s_{um}/kD = normalized shear strength of clay;
t = thickness of steel anchor;
T_c = overlying clay thickness;
T_c/D = overlying clay thickness ratio;
V = vertical load;
V_{max} = maximum vertical pullout capacity;
V_{ult} = ultimate vertical pullout capacity;
V_0 = vertical pullout capacity;
$V_0/\gamma' D^3$ = dimensionless vertical pullout capacity;
Z_{cl} = centerline loading depth;
Z_{cl}/L = loading depth of the normalized centerline;
$Z_{cl,opt}$ = loading depth of the optimum centerline;
$Z_{cl,opt}/L$ = normalized loading depth of the optimum centerline;
Z_{pad} = depth of the padeye;
Z_{pad}/L = normalized padeye depth;
$Z_{pad,opt}$ = optimum depth of the padeye;
$Z_{pad,opt}/L$ = normalized optimum depth of the padeye;
γ' = effective unit weight;
θ = loading angle of the mooring line;
θ_{max} = loading angle where V_{max} occurred;
μ = friction coefficient;
υ = Poisson's ratio;
ϕ = friction angle; and
ψ = dilatancy angle.

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